



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

LAB ASSIGNMENT-3

Submitted by:

Ryan Kapoor

1024150188

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1. Write a program to implement

(a) pointer to an object

(b) this pointer.

Practice both . (dot operator) and -> (arrow operator).

OOPS > A3 > Q1.cpp > Demo > Demo(int)

```
1  #include<iostream>
2  using namespace std;
3  class Demo{
4      int x;
5      public:
6          Demo(int x){
7              this->x=x;
8          }
9          void show(){
10             cout<<this->x<<"\n";
11         }
12 };
13 int main(){
14     Demo obj(10);
15     Demo *p=&obj;
16     obj.show();
17     p->show();
18 }
19
```

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2. Write a program to swap private values of two classes using a friend function.

```

OOPS > A3 > G+ Q2.cpp > ...
1  #include<iostream>
2  using namespace std;
3  class B;
4  class A{
5      int x;
6      public:
7          A(int x){
8              this->x=x;
9          }
10         friend void swap(A&,B&);
11         void show(){
12             cout<<x<<" ";
13         }
14     };
15     class B{
16         int y;
17         public:
18             B(int y){
19                 this->y=y;
20             }
21             friend void swap(A&,B&);
22             void show(){
23                 cout<<y<<" ";
24             }
25     };
26     void swap(A &a,B &b){
27         int t=a.x;
28         a.x=b.y;
29         b.y=t;
30     }
31     int main(){
32         A a(10);
33         B b(20);
34         swap(a,b);
35         a.show();
36         b.show();
37     }
38

```

```

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```

3. Write a program to add data objects of two different classes using friend functions.

```

OOPS > A3 > Q3.cpp > ...
1  #include<iostream>
2  using namespace std;
3  class B;
4  class A{
5      int x;
6      public:
7          A(int x){
8              this->x=x;
9          }
10         friend int add(A,B);
11     };
12     class B{
13         int y;
14         public:
15             B(int y){
16                 this->y=y;
17             }
18             friend int add(A,B);
19     };
20     int add(A a,B b){
21         return a.x+b.y;
22     }
23     int main(){
24         A a(10);
25         B b(20);
26         cout<<add(a,b);
27     }

```

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4. Write a program to demonstrate the working of friend class.

```
OOPS > A3 > Q4.cpp > ...
1  #include<iostream>
2  using namespace std;
3  class A{
4      int x;
5      friend class B;
6      public:
7          A(int x){
8              this->x=x;
9          }
10 };
11 class B{
12     public:
13         void show(A a){
14             cout<<a.x;
15         }
16 };
17 int main(){
18     A a(50);
19     B b;
20     b.show(a);
21 }
22
```

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5. Write a program using Array of Objects to display area of multiple rectangles.

```

OOPS > A3 > Q5.cpp > Rect > area()
1  #include<iostream>
2  using namespace std;
3  class Rect{
4      int l,b;
5      public:
6          void set(int l,int b){
7              this->l=l;
8              this->b=b;
9          }
10         int area(){
11             return l*b;
12         }
13     };
14     int main(){
15         Rect r[3];
16         r[0].set(2,3);
17         r[1].set(4,5);
18         r[2].set(6,7);
19         for(int i=0;i<3;i++)
20             cout<<r[i].area()<<" ";
21     }
22
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6 20 42
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```

6. Write a program to define function cube() as inline for calculating cube of a number.

```

OOPS > A3 > Q6.cpp > main()
1  #include<iostream>
2  using namespace std;
3  inline int cube(int x){
4      return x*x*x;
5  }
6  int main(){
7      int n;
8      cin>>n;
9      cout<<cube(n);
10 }
11

```

7. Write a program to pass an object as an argument and return the object from a function.
- a. Use pass-by-value
 - b. Use pass-by-address

```
OOPS > A3 > q7.cpp > ...
1  #include<iostream>
2  using namespace std;
3  class A{
4      public:
5          int x;
6          A(int x=0){
7              this->x=x;
8          }
9  };
10 A modify(A a){
11     a.x+=10;
12     return a;
13 }
14 int main(){
15     A a(5);
16     A b=modify(a);
17     cout<<b.x;
18 }
19
```

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